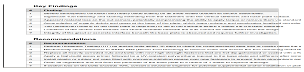

Image Analysis Report

Generated	2026-04-24 07:30:05Z
Source	corrosion.jpg
Report ID	3d1020cb-6fa0-4e70-8983-d851c527843e

Executive Summary

The image exhibits severe localized corrosion and significant coating failure, particularly concentrated along a horizontal seam. Given the visible metal delamination and scaling, the severity is high, posing a risk to structural integrity or pressure containment.

Key Findings

#	Finding
1	Severe localized corrosion in the center with visible material delamination and 'rust scabbing'.
2	Preferential corrosion attack along a horizontal feature, likely a weld seam or lap joint.
3	Widespread coating breakdown leading to secondary atmospheric corrosion and rust staining.
4	Presence of heavy, dark iron oxide scale suggests a prolonged period of moisture retention and oxidation.
5	Surface pitting is suspected under the central scaled area but cannot be quantified without cleaning.
6	Visible vertical rust streaks indicating active corrosion run-off from the primary site.

Localized detections (VLM, approximate)

- Severe Scaling and Metal Loss (conf. 95%): x=0.350, y=0.150, w=0.380, h=0.650
- Seam Corrosion (conf. 88%): x=0.000, y=0.450, w=1.000, h=0.120

Recommendations

#	Recommendation
1	Perform immediate Ultrasonic Thickness (UT) gauging at the center of the defect to assess remaining wall thickness.
2	Abrasive blast clean the area to SSPC-SP 10 (Near-White Metal) to fully expose the extent of metal loss.
3	Evaluate for a weld-on repair patch or structural reinforcement if wall loss exceeds the allowable limit (typically 12.5-20% depending on code).
4	Perform Dye Penetrant Inspection (DPI) or Magnetic Particle Inspection (MPI) on the horizontal weld seam to check for surface-breaking cracks.
5	Apply a high-build epoxy coating system specifically formulated for salt/industrial environments after surface preparation.
6	Establish a localized monitoring point and re-inspect within 6 months to ensure the repair/coating remains intact.
7	Ensure all personnel maintain a safe distance until the integrity of the component (if pressurized) is verified.